Домашнее задание 11.

Задача 1.

$n=15, \quad d=5, \quad GF(2^4)$

$$g(x) = (x-\alpha)(x-\alpha^2)(x-\alpha^3)(x-\alpha^4) = (x^2-\alpha^2x-\alpha x+\alpha^3)(x^2-\alpha^4x-\alpha^3x+\alpha^7) = (x^2-\alpha^5x+\alpha^3)\cdot(x^2-\alpha^7x+\alpha^7) = x^4-\alpha^7x^3+\alpha^7x^2-\alpha^5x^3+\alpha^{12}x^2-\alpha^{12}x+\alpha^3x^2-\alpha^{10}x+\alpha^{10} =$$

$$= x^4-\alpha^{13}x^3+\alpha^6x^2+\alpha^3x+\alpha^{10}$$

$$g(x) = x^4+\alpha^{13}x^3+\alpha^6x^2+\alpha^3x+\alpha^{10}$$

Задача 2.

$РС(15,9), \quad GF(2^4), \quad d=7$

$$p(x) = 1+x+x^4$$

$$v(x) = \alpha+\alpha^3x+\alpha^{10}x^2+\alpha^{12}x^3+\alpha^{13}x^4+\alpha^3x^5+\alpha^9x^6+\alpha^5x^7+x^8+\alpha^{10}x^9+\alpha^8x^{10}+$$

$+\alpha x^{11} + \alpha x^{12} + \alpha^{13} x^{13} + \alpha^{2} x^{14}$

$$v = (\alpha, \alpha^3, \alpha^{10}, \alpha^{12}, \alpha^{13}, \alpha^3, \alpha^9, \alpha^5, 1, \alpha^{10}, \alpha^8, \alpha, \alpha, \alpha^{13}, \alpha^2)$$

$$S_i = \sum_{j=0}^{14} v_j (\alpha^i)^j$$

$$S_1 = \alpha + \alpha^3 \alpha + \alpha^{10}\cdot\alpha^2 + \alpha^{12}\cdot\alpha^3 + \alpha^{13}\cdot\alpha^4 + \alpha^3\cdot\alpha^5 + \alpha^9\cdot\alpha^6 + \alpha^5\cdot\alpha^7 + \alpha^8 + \alpha^{10}\cdot\alpha^9 + \alpha^8\cdot\alpha^{10} + \alpha\cdot\alpha^{11} + \alpha\cdot\alpha^{12} + \alpha^{13}\cdot\alpha^{13} + \alpha^2\cdot\alpha^{14} =$$

$$= \cancel{\alpha^1} + \cancel{\alpha^4} + \cancel{\alpha^{12}} + \cancel{\alpha^0} + \alpha^2 + \cancel{\alpha^8} + \cancel{\alpha^0} + \alpha^{12} + \cancel{\alpha^8} + \cancel{\alpha^4} + \alpha^3 + \cancel{\alpha^{12}} + \alpha^{13} + \alpha^{11} + \cancel{\alpha^1} =$$

$$= \cancel{x^2} + \cancel{1} + \cancel{x} + \cancel{x^2} + \cancel{x^3} + \cancel{x^3} + \cancel{1} + x^2 + x^3 + \cancel{x} + x^2 + x^3 = \varnothing$$

аналогично:

$$S_2 = S_3 = S_4 = S_5 = S_6 = \varnothing$$

=> принятое слово не содержит ошибок

$$e(x) = \varnothing$$